Towards a novel benchmark for RNA design

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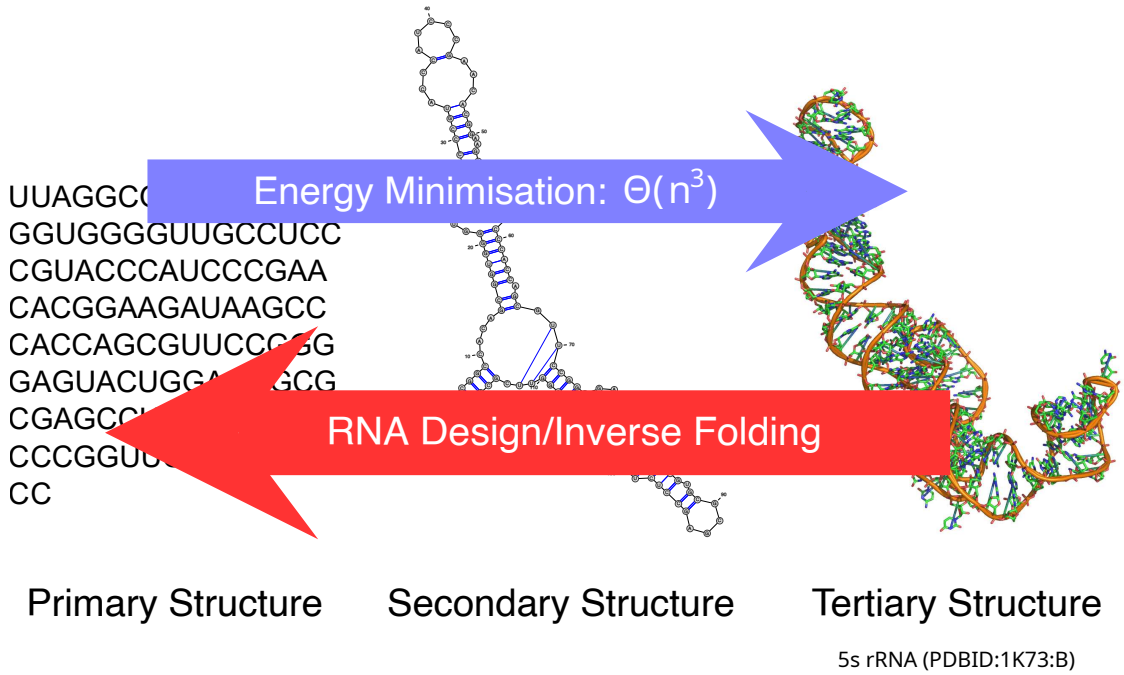

Motivation. Ribonucleic Acids (RNAs) are versatile biomolecules that act both in nature as a medium to store genetic information (RNA viruses), and as an enzyme that participates in a rich variety of cellular functions (replication, regulation, direct sensing of metabolites...). Recent progress towards understanding the interplay between RNA sequence, structure and function has led to novel RNA-based therapeutics and gene-editing technologies (CRISPR/Cas9). Such applications require the development of computational methods for a rational design of synthetic RNAs, built to ensure both their affinity towards a targeted function, and the absence of undesired interactions with natural biomolecules ($\approx$ absence of side effects).

Problem formalization. RNAs can be abstracted as a sequence ω, composed of individual elements called nucleotides: Adenine (A), Cytosine (C), Uracile (U) and Guanine (G). Upon their synthesis, RNA folds and adopts complex structures, mediated by hydrogen bonds. Those result in the formation of a set of base pairs (BPs) $S = \{(i, j)\}$, each denoting the interaction of nucleotides at positions i and j. The set of BPs in an RNA structure is called its secondary structure, and each sequence/structure pair (ω, S) can be associated with a free-energy $E(\omega, S)$, quantifying how stable the structure is over this sequence. The structure $S^\star(\omega) := \text{argmin}_S\, E(\omega, S)$ is called the minimum free-energy (MFE) secondary structure of ω. In this context, a popular formalization of RNA design, called inverse folding problem, aims at finding an RNA sequence whose MFE structure matches a predefined target structure T.

Problem 1 (Inverse folding):
Input: Target RNA secondary structure T, length n
Output: An RNA sequence $\omega \in \{A, C, G, U\}^n$ such that $E(\omega, S) > E(\omega, T), \forall S \neq T$

Given the NP-hardness of the problem, many research groups have proposed algorithms for the inverse folding problem, based on (meta)-heuristics, local search, machine learning or exponential backtracking approaches. However, evaluating the relative merits, and scope of validity, of the resulting design tools remains highly challenging due to the absence of a comprehensive benchmark, focusing on difficult instances. For instance, the popular ETERNA benchmark, consisting of 100 instances selected to be challenging to online Human players, is now overwhelmingly solved by top-of-the-shelf methods [1].

Objective. The goal of this Bachelor's thesis is to generate a benchmark of "hard" instances for the RNA Inverse Folding problem, and contribute an objective assessment of the success and limits of current state-of-the-art methods. Towards that goal, the successful candidate will build on the experience acquired by the mentoring team, including the prior identification of some critical parameters and properties that make RNA design overly easy. For instance, overly large individual helices tend to make the design task largely trivial, and so is a restriction on low degrees for multiple loops. In addition, recent work have revealed the existence of a set of relatively small structural motifs associated with a surprisingly-limited number of local designs (*i.e.* such motifs/structures admit a more stable alternative structure for a large majority of nucleotide assignments). Towards creating challenging instances, these motifs can be added and combined into large structures for which the number of designs can be bounded (*needle in a haystack*). We expect the intern to propose and implement an algorithm to combine these motifs and evaluate an upper bound on the number of sequences that could be a potential design for the target structure. The intern will also evaluate the performances of state-of-the-art methods on the contributed benchmark. Reference methods include methods based on local search (RNAInverse [2, 1], evolutionary computing (DesiRNA [3]; eM2dRNAs [4]), NUPACK [5]), Monte Carlo search (NEMO [6]), and Reinforcement Learning (Learna [7]). Overall, a successful execution of this internship will provide an evidentiary basis for the development of future RNA design methodologies.

Profile. The successful candidate is a student at the Bachelor of Ecole Polytechnique, equipped with a comprehensive background in Computer science:

- They should have general knowledge in algorithm design. Specific experience in complexity, graphs, dynamic programming or combinatorics will be a plus but not mandatory;
- They should have general interest in Molecular Biology, but a formal education in this field is not mandatory, as the required background in Biology and/or RNA will be presented at the start of the internship;
- They should be fluent in Python Programming. Prior experience with reproducible research, including workflow management systems (*e.g.* SnakeMake), or in accessing remote computational resources (clusters) will be a plus.

References

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